

July 31, 2025

Dear Professor,

We are pleased to submit our manuscript entitled "*Quantitative Assessment of Performance Degradation in Model-Based Galileo SAR Systems Considering Human Factors: Hypothermia-Induced Physiological and Psychological Variability*" for consideration in Journal of Internet of Things.

This work addresses a critical issue in the operational reliability of the European Galileo satellite system, particularly in its mission for Search and Rescue (SAR). Our study aligns with the scope of *Engineering Cyber-Physical Human Systems*, as it integrates human factors, specifically, the impact of hypothermia on physiological and psychological performance, into the modeling and analysis of space-based SAR operations.

Key contributions of our paper include:

1. **Model-Based System Representation:** We construct a detailed model that accurately captures various degradation modes of the Galileo SAR system, collected from official documentation.
2. **Human Status Modeling:** The SAR mission model incorporates clinically-sourced parameters to build a stage-based stochastic model representing human physiological and cognitive degradation due to hypothermia.
3. **Compositional Modeling and Verification:** The composed model is analyzed in two phases. The first simulates system evolution over time, while the second uses a game-theoretic approach to model the instinctive human behavior related to survival and service attempts.

Our analysis offers a comprehensive evaluation of the SAR system's performance and dependability, grounded in a set of rigorously defined parameters. We demonstrate the strengths of Continuous-Time Markov Chains (CTMCs) and Concurrent Stochastic Games (CSGs) in accurately capturing human-state-driven variations in mission outcomes.

We believe this research will provide valuable insights for the Engineering Cyber-Physical Human Systems community and support the development of more resilient, human-aware SAR systems.

To facilitate reproducibility and further research, we have made our PRISM models and source code publicly available via GitHub. The repository includes detailed deployment instructions to support easy integration and experimentation.

Thank you for considering our submission. We look forward to your response.

Sincerly,

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